Module 6: Introduction to Generative Adversarial Networks (GANs)

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Outline of this Module

1. Introduction to Generative Models
2. Basics of Generative models
3. Introduction to GANs

Introduction to Generative Models

Discriminative models and Generative models

What are Generative models

Definition

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- Can be used to **generate new data samples** that are not present in the original dataset.
- Often learns a compressed and meaningful **latent space representation** of the data.

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- **Missing Data Imputation:** Fill in gaps in incomplete data.
- **Anomaly Detection:** Identify samples that deviate from the learned distribution.
- **Representation Learning:** Discover meaningful low-dimensional representations of the data.

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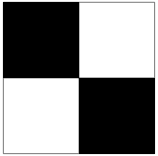

Example Image

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- Imagine we have a dataset of 2x2 images.
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- In this discrete case, we can simply count the occurrences of each of the 16 possible images and normalize to get probabilities.

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  - ... and so on for all 16 states.
- For real images, the number of pixels is huge ($256 \times 256 \times 3$ for color images), making explicit enumeration impossible. Generative models use clever techniques to approximate this high-dimensional distribution.

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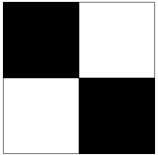

Sampled Image 1

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Hypothetical Joint Probabilities:

- $P(x_{11} = 1, x_{12} = 0, x_{21} = 0, x_{22} = 0) = 0.05$
- $P(x_{11} = 1, x_{12} = 0, x_{21} = 0, x_{22} = 1) = 0.10$
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- If we have $P(x_1, x_2, x_3, x_4)$ for our 2x2 grid, we can find the probability of $P(x_1)$ (e.g., the top-left pixel being black) by summing over all possible states of x_2, x_3, x_4:

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- This tells us the overall likelihood of a specific pixel value, irrespective of other pixels.

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- Using Bayes' theorem and the learned joint distribution:

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- For our 2x2 grid, if we know the top-left pixel is black and others are unknown, the model can predict the most likely full image.

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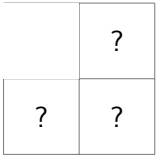

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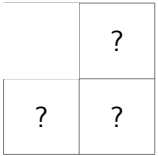

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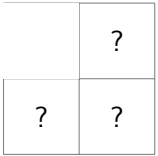

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- From our list, $P(1,0,1,0) = 0.15$ is the maximum.

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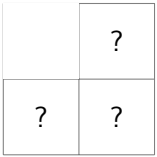
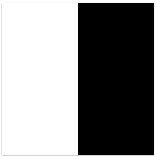


Input (Partial) $\xrightarrow{\text{Generative Model}}$ Output (In-painted)



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- The model fills in the '?' based on the learned distribution, making the most probable completion.

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- This is computationally impossible for explicit generative models.

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  - **Directly predict:** A neural network (e.g., U-Net in Diffusion Models) is trained to directly predict the missing pixels given the context.

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  - **Directly predict:** A neural network (e.g., U-Net in Diffusion Models) is trained to directly predict the missing pixels given the context.
  - **Latent space manipulation:** For VAEs and GANs, in-painting can involve encoding the partial image into a latent space, manipulating the latent representation, and decoding it.

How Modern Models Address the Bottleneck

- Modern generative models (GANs, VAEs, Diffusion Models) do not explicitly learn or store the full joint probability table.
- Instead, they learn a function that can approximate this distribution or generate samples from it.
- For in-painting, they use these learned functions to:
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  - **Latent space manipulation:** For VAEs and GANs, in-painting can involve encoding the partial image into a latent space, manipulating the latent representation, and decoding it.
- These methods avoid the explicit enumeration of all possible pixel combinations, making in-painting feasible for high-resolution images.

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- This is known as a generative classifier.

Generative vs. Discriminative Classifiers

- Generative Classifier: Models $P(\mathbf{x}|y)$ and $P(y)$. Can generate data.
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- Generative classifiers can be more robust to missing data and provide richer information, but often require more data to learn $P(\mathbf{x}|y)$ accurately.

Illustrations of Generative Models

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Text-to-Audio

Meta's AudioGen: A state-of-the-art model that can generate high-fidelity audio samples, including sound effects and simple musical pieces, directly from text descriptions.

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- The training process aims to find the optimal parameters θ_g^* such that the distribution of $G(\mathbf{z})$ matches the distribution of real data.

Illustrative example of Generative model

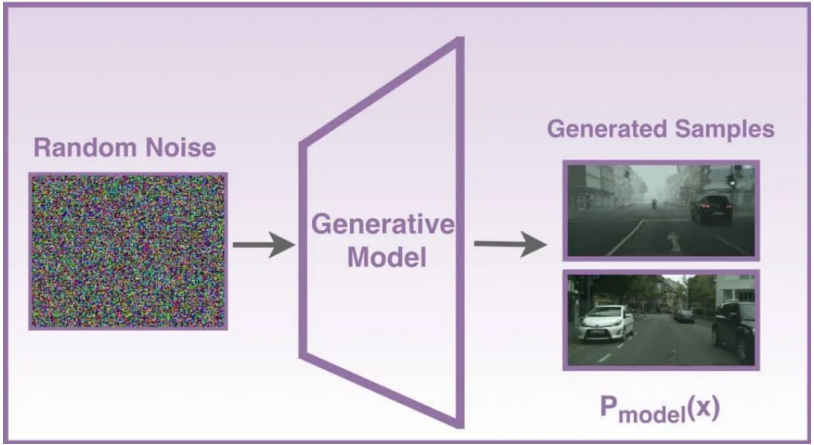

Figure: Image visualising the concept of diverse image generation from noise. (source: learnopencv.com)

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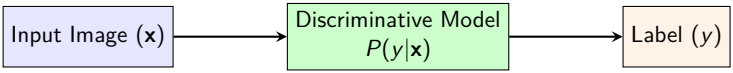

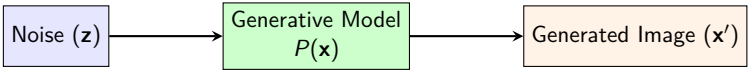

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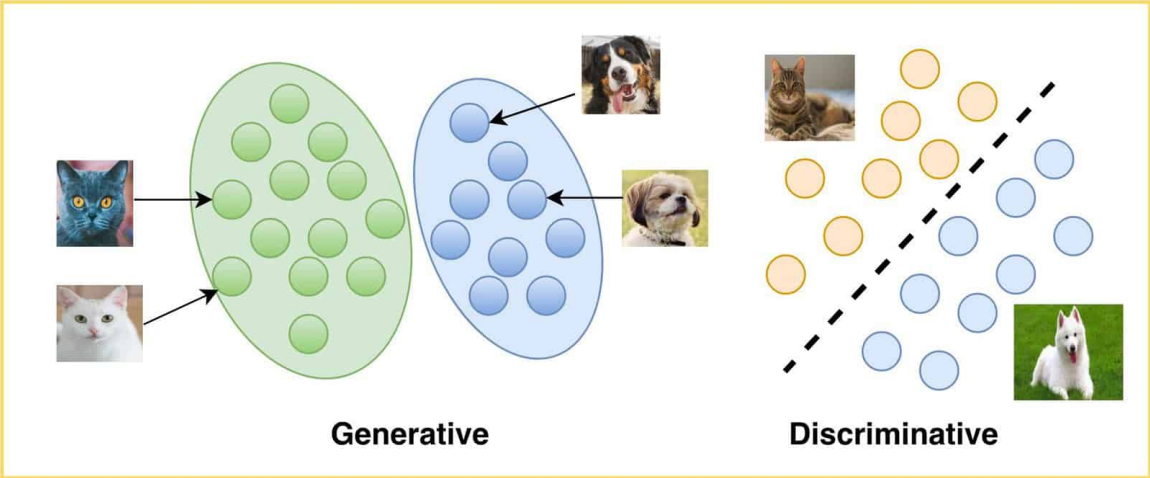

Figure: Different Learning patterns of Discriminative and Generative models.(source: learnopencv.com

Basics of Generative models

Latent Space, Noise Sampling and Types of Generative models

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Other Key Pre-Deep Learning Models

Factor Analysis (FA)

Idea: Assumes observed data is a linear combination of unobserved "factors" (latent variables) plus noise. Tries to find causal structure.
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Matrix Factorization (e.g., SVD)

Idea: Decomposes a large matrix (e.g., user-item ratings) into smaller matrices of latent factors.

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The Problem: The latent space was often unstructured. Picking a random point z and decoding it would usually produce garbage, not a valid new data point. The focus was on **inference**, not **generation**.

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This was enabled by deep learning models like **Variational Autoencoders (VAEs)** and **Generative Adversarial Networks (GANs)**, which learned to create a structured, continuous latent space.

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   - The "noise" comes from the randomness of this direct sampling process. Each sample is a random draw according to the learned probabilities. There's no separate "latent vector" input.

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- The "noise" is the source of variety. Different z vectors map to different generated outputs, allowing the model to produce diverse samples from the learned data manifold.

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Analogy

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- **Deep Model:** You pick a random set of DNA instructions (z), and then an organism grows from it (the Generator). Different DNA leads to different organisms.

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- Key types include:
  - Variational Autoencoders (VAEs)
  - Generative Adversarial Networks (GANs)
  - Autoregressive Models (e.g., PixelRNN, WaveNet)
  - Flow-based Models
  - Diffusion Models (currently state-of-the-art for image generation)

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- **Solution:** Instead of trying to directly sample from $P(\mathbf{x})$ in pixel space, let's learn a *mapping* from a simpler, continuous distribution (the latent noise z) to the data manifold.

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3. But we can pick a random point in a simpler, smaller "idea space" (the latent space Z).

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5. Each unique z vector acts as a unique "seed" or "recipe" for a specific image, ensuring variety while staying on the data manifold.

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- Diffusion Models emerged more recently as another powerful approach using structured noise.

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Question: "What are the hidden factors in my data?"	**Question:** "How can I create new data that looks like my data?"

Table: The paradigm shift from data analysis to data synthesis.

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- **Diffusion / Score-Based Models:** Systematically add noise to data, then learn to reverse the process. State-of-the-art for images. (*Examples: DDPM, Stable Diffusion, DALL-E 2*)

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- **Generator:** Creates fake data from random noise. Tries to fool the discriminator.
- **Discriminator:** Tries to distinguish real data from fake data.
- Through this adversarial process, the generator learns the true data distribution implicitly.

Examples: DCGAN, StyleGAN, CycleGAN

Taxonomy Summary Table

Category	Sub-Category	Key Idea	Examples
Explicit Density	**Tractable**	Directly compute $p(x)$	PixelCNN, Glow
	Approximate	Optimize a proxy for $p(x)$	VAEs, Diffusion Models
Implicit Density	-	Learn to sample from $p(x)$	GANs, StyleGAN

- **GANs** are known for sharp, high-fidelity samples but can be unstable to train.
- **VAEs** are stable and have a well-behaved latent space but can produce blurry results.
- **Flows** offer exact likelihoods but can be computationally expensive.
- **Diffusion Models** are the current leaders in sample quality and diversity but are slow to sample.

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- The modern generative era began with a conceptual shift: using **random noise** as an input to sample from a learned, structured latent space.
- Today's powerful generative models (GANs, VAEs, Diffusion) are the direct descendants of this idea, each with a unique approach to learning the underlying data distribution.

Thank you!

Introduction to GANs

The GAN architecture, Training and Challenges

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The Creative Competition

The game ends when the artist's forgeries are so good that the critic is fooled about half the time.

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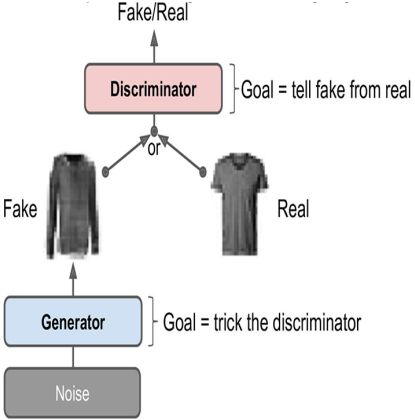

Figure: GAN architecture.

Source: Hands-On Machine Learning with Scikit-Learn, Keras TensorFlow.

The Models and their Goals

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$$\max_{D} \left(\underbrace{\mathbb{E}_{\mathbf{x}\sim p_{data}(\mathbf{x})}[\log D(\mathbf{x})]}_{\text{Correctly label real as real}} + \underbrace{\mathbb{E}_{\mathbf{z}\sim p_{\mathbf{z}}(\mathbf{z})}[\log(1 - D(G(\mathbf{z})))]}_{\text{Correctly label fake as fake}} \right)$$

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Its objective is to **minimize** the second term of the Discriminator's objective:

$$\min_G \left(\mathbb{E}_{\mathbf{z} \sim p_{\mathbf{z}}(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))] \right)$$

This is equivalent to training G to maximize $\log(D(G(\mathbf{z})))$, pushing the Discriminator's output for fake images towards 1.

GAN Training Objective

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The full training objective pits the Generator against the Discriminator. The Generator tries to minimize the value function that the Discriminator tries to maximize:

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The Goal: Nash Equilibrium

The ideal outcome of this game is to find a **Nash Equilibrium**. At this point, the Generator produces perfectly realistic data, and the Discriminator is unable to distinguish real from fake, outputting a 50% probability for all inputs.

GAN Training Algorithm: Phase 1 - Train the Discriminator

Objective: Teach D to distinguish real from fake

1. **Sample data:**
   - Real images: $\mathbf{x} \sim p_{\text{data}}$

from dataset

Real $\mathbf{x}$

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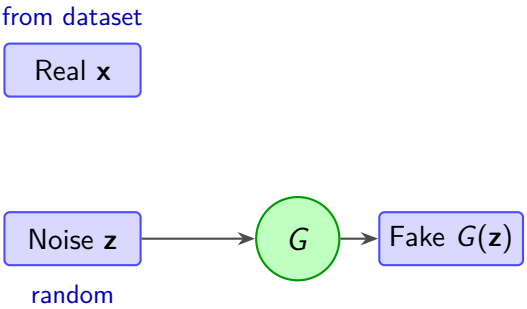

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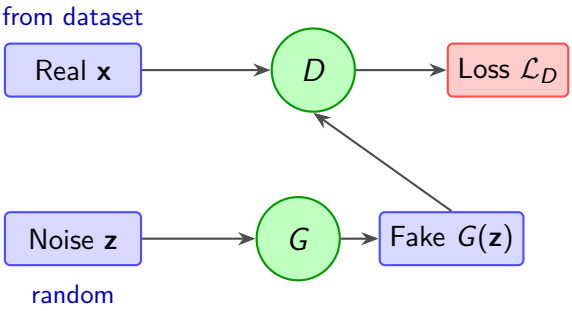

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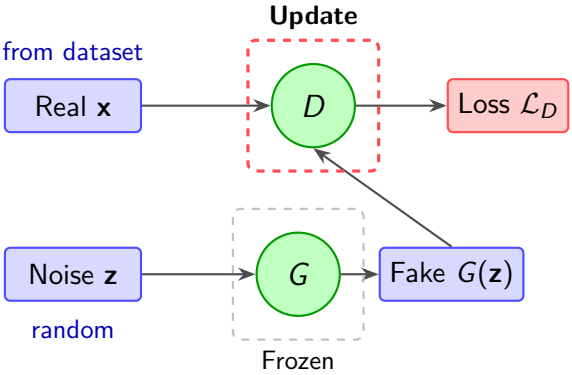

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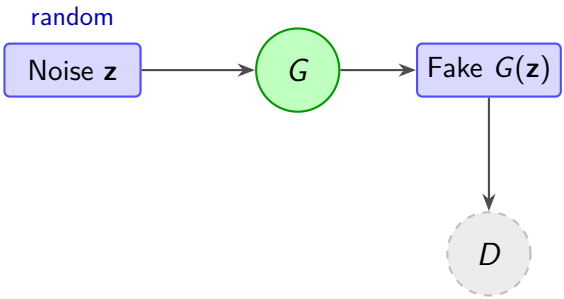

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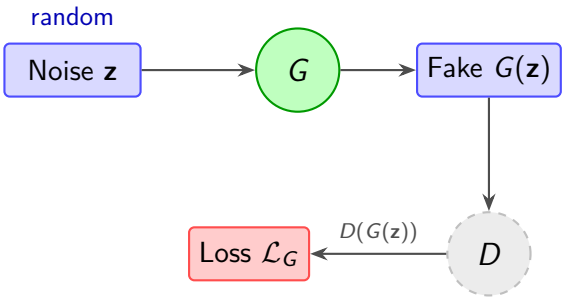

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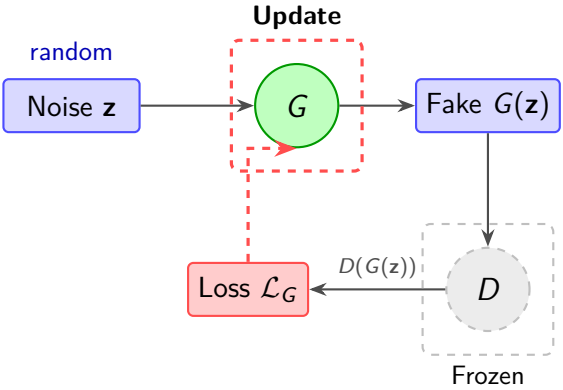
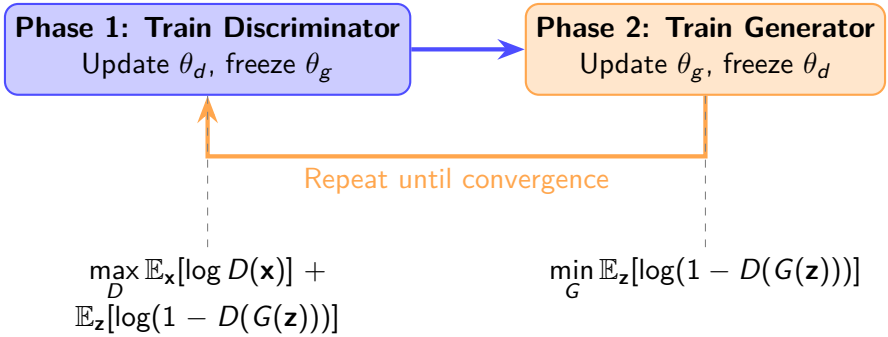


Result: G improves at generation

(Gradients flow back through D to update G)

GAN Training: Alternating Optimization



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- **Activation Functions:**
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  - Use **LeakyReLU** in the discriminator for all layers.

DCGAN generations

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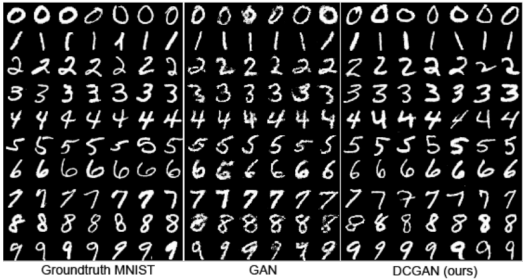

Figure: Qualitative Comparison between GAN and DCGAN

(source: "UNSUPERVISED REPRESENTATION LEARNING WITH DEEP CONVOLUTIONAL GENERATIVE ADVERSARIAL NETWORKS)

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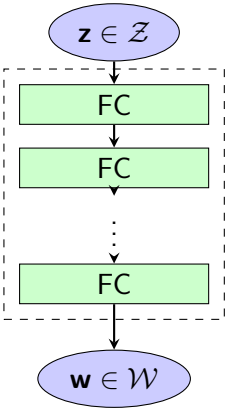

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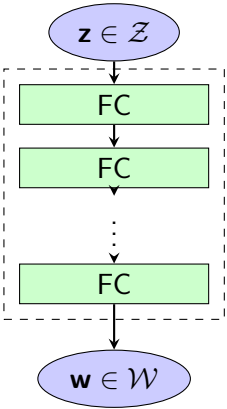

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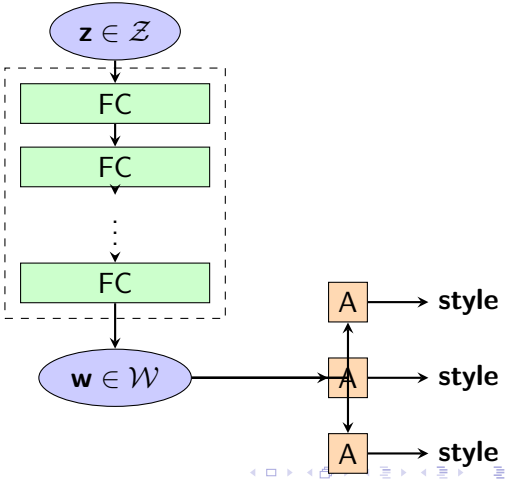
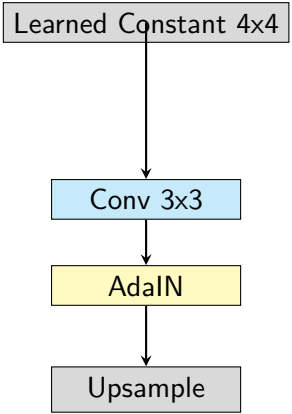
- This disentangles the latent space and reduces correlations.
- From **w**, multiple **style vectors** are produced via affine transformations. These control different visual features (e.g., hair color, face shape) at different scales.



StyleGANs

2. Synthesis Network

- Responsible for generating the image, starting from a learned constant input.



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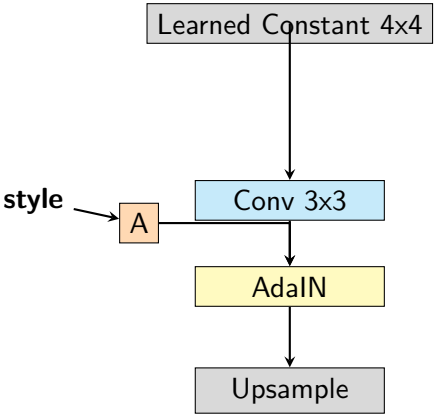
- **AdaIN (Adaptive Instance Normalization):** Uses the style vectors from the mapping network to control the scale and bias of features at each convolutional layer.



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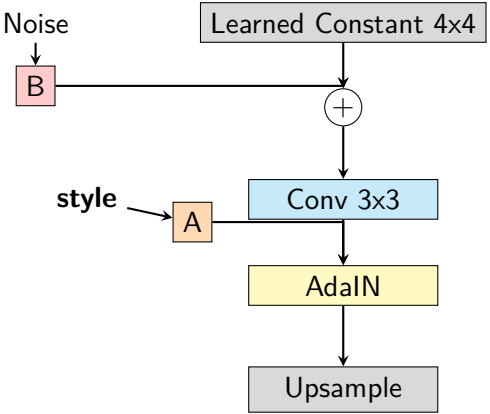
- **AdaIN (Adaptive Instance Normalization):** Uses the style vectors from the mapping network to control the scale and bias of features at each convolutional layer.
- **Stochastic Variation:** Adds explicit noise at different layers to model random features like freckles or hair placement, which is more efficient than encoding this in the latent code **z**.



StyleGAN Architecture

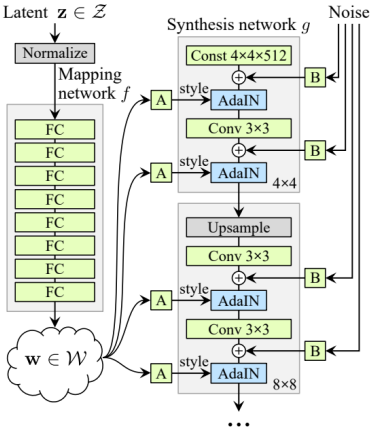


Figure: StyleGAN generator Architecture.
(source: "A Style-Based Generator Architecture for Generative Adversarial Networks")

StyleGAN Illustration

Figure: StyleGAN Face Generation.
(source: "A Style-Based Generator Architecture for Generative Adversarial Networks")

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- The GAN may cycle through a few modes, never learning the full data distribution.

Mode Collapse Illustration

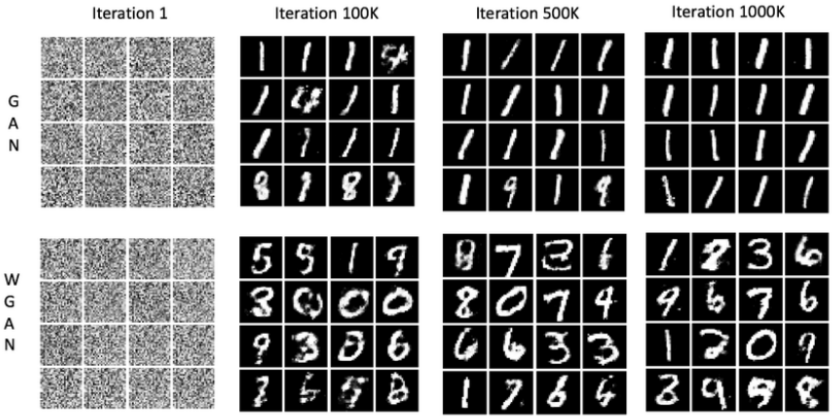

Figure: Illustration of Mode Collapse in Traditional GAN, WGAN aims to reduce mode collapse. *(source: medium)*

Training Challenges in GANs

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- **Vanishing Gradients:** If the discriminator gets too good too quickly, the generator's gradients can vanish, causing learning to stop.
- **Hyperparameter Sensitivity:** GANs are very sensitive to the choice of hyperparameters, requiring significant fine-tuning.

Plausible solutions

Researchers have proposed many techniques to stabilize training and avoid mode collapse.

Techniques to Improve Stability and Diversity

- **Wasserstein GAN (WGAN)**: It measures the **Wasserstein distance** between the real and generated data distributions. By minimizing the distance between the entire distributions, the generator is strongly incentivized to produce a wide variety of outputs to match the diversity of the real data, thus significantly reducing mode collapse.

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- **Progressive Growing of GANs:** Start by generating very small images (e.g., 4×4) and progressively add layers to both networks to generate larger images ($8 \times 8, 16 \times 16$, etc.). This simplifies the learning task at each stage.

Key Takeaways

- **Adversarial Training:** GANs consist of a Generator and a Discriminator locked in a competitive, zero-sum game, which drives the generator to produce increasingly realistic data.
- **Powerful Generative Models:** They are capable of generating very crisp and realistic images compared to other generative models.
- **Architecture Evolution:** The field has progressed rapidly from the original GAN to architectures like DCGANs, which established stable convolutional designs, and StyleGANs, which provide fine-grained control over the synthesis process.
- **Training is Challenging:** Issues like **mode collapse** and **training instability** are significant hurdles, though many techniques exist to mitigate them.
- **Meaningful Latent Space:** GANs can learn rich, meaningful latent representations that allow for semantic operations, such as vector arithmetic on faces.